Second Problem Sheet
on
Analytical Mechanics

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Topic: Lagrangian Mechanics

"The reader will find no figures in this work. The methods which I set forth do not require either constructions or geometrical or mechanical reasonings: but only algebraic operations, subject to a regular and uniform rule of procedure."

-Joseph-Louis Lagrange (talking about his work Mécanique Analytique)

Problem 1

In the following exercises, show that the principle of virtual work holds and use this fact to find the equations of motion.

i) A block of mass m can slide freely on a wedge that is moving in a specified manner on a horizontal surface; that is, the coordinate X shown in Figure 1 is assumed to be a given function of time.

ii) Consider a bead of mass m that can slide along a rod which rotates about a vertical axis with constant angular velocity ω. The rod makes a constant angle θ_0 with the vertical axis (see Figure 2)

iii) A simple pendulum of mass m_2 hangs from a block of mass m_1 which is attached to a spring of stiffness k. The block lies on a horizontal surface without friction.

iv) A bead of mass m can slide freely along a hoop of radius a. The hoop rotates with constant angular velocity ω about a fixed point of the hoop on a horizontal plane (see Figure 4).

v) A block of mass m_2 is tied to a wall by means of a rope that passes over a pulley attached to a wedge of mass m_1. The block lies on the wedge, which is free to move on a horizontal surface (see Figure 5). Neglect the masses of the rope and the pulley, and the friction forces.

Problem 2

Find the Lagrangian of each system in the previous problem.

Problem 3

If L is a Lagrangian for a system of n degrees of freedom satisfying Lagrange's equations, show that

$$L' = L + \frac{\mathrm{d}F(q_i, t)}{\mathrm{d}t}$$

also satisfies Lagrange's equations where F is any arbitrary, but differentiable, function of its arguments. This is sometimes known as a **gauge transformation**.

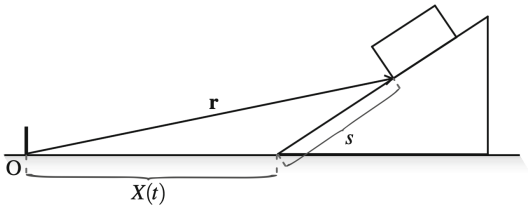


Figure 1: The position of the wedge is given by $X(t)$

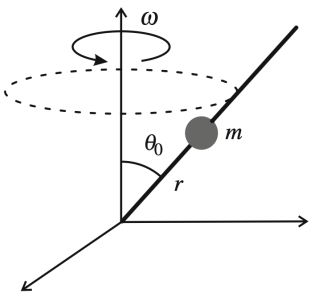


Figure 2: The bead can slide freely along the rod

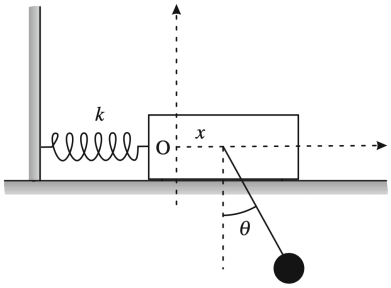


Figure 3: The bead can slide freely along the rod

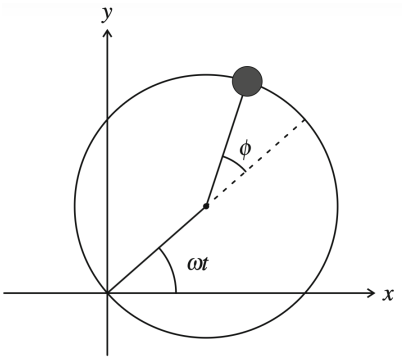


Figure 4: The bead can slide freely along the rod

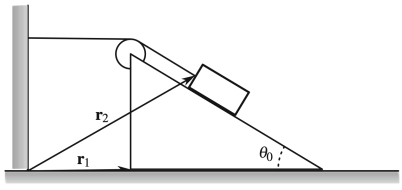


Figure 5: The bead can slide freely along the rod

Problem 4

Show that Lagrange's equations can also be written as

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \dot{T}}{\partial \dot{q}_j} - 2\frac{\partial T}{\partial q_j} = Q_j.$$

These are sometimes known as the **Nielsen** form of the Lagrange's equations

Problem 5

The electromagnetic field is invariant under a Gauge transformation of the scalar and vector potential given by

$$\mathbf{A} \to \mathbf{A} + \nabla\psi(\mathbf{r}, t)$$
$$\phi \to \phi - \frac{1}{c}\frac{\partial \psi}{\partial t},$$

where ψ is any differentiable function. What effect does this gauge transformation have on the Lagrangian of a particle moving in the electromagnetic field? Is the motion affected?

Problem 6

Let $q_1, \ldots, q_n$ be a set of independent generalized coordinates for a system of n degrees of freedom, with a Lagrangian $L(q, \dot{q}, t)$. Suppose we transform to another set of independent coordinates $s_1, \ldots, s_n$ by means of transformation equations

$$q_i = q_i(s_1, \ldots, s_n; t), \quad i = 1, \ldots, n.$$

(Such a transformation is called a **point transformation**.) Show that if the Lagrangian function is expressed as a function of $s_j, \dot{s}_j$, and t though the equations of transformation, then L satisfies Lagrange's equations with respect to the s coordinates:

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{s}_j} - \frac{\partial L}{\partial s_j} = 0.$$

Problem 7

A Lagrangian for a particular physical system can be written as

$$L' = \frac{1}{2}m\left(a\dot{x}^2 + 2b\dot{x}\dot{y} + c\dot{y}^2\right) - \frac{1}{2}K\left(ax^2 + 2bxy + cy^2\right),$$

where a, b, and c are arbitrary constants but subject to the condition that $b^2 - ac \neq 0$. Find the equations of motion and examine particularly the two cases $a = 0 = c$ and $b = 0, c = -a$. What is the physical system described by the above Lagrangian? Show that the usual Lagrangian for this system is related to L' by a point transformation (cf. Problem 4). What is the significance of the condition on the value of $b^2 - ac$?

Problem 8

Write down the Lagrangian for a one-dimensional harmonic oscillator. Show that the Lagrangian is invariant under the transformations

i) $\psi_\epsilon(x) = x + \epsilon \sin \omega t$
ii) $\xi_\epsilon(x) = x + \epsilon \cos \omega t$

with $\omega = \sqrt{k/m}$ and use Noether's theorem to obtain the correspondent conserved quantity. Furthermore, write the amplitude A in terms of the conserved quantities and thus conclude that A is a conserved quantity.